

Craig M. Weeks

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EDUCATION

Carnegie Mellon University

Doctor of Philosophy in Mechanical Engineering | GPA: 4.0/4.0

Master of Science in Mechanical Engineering

Pittsburgh, PA

August 2026

May 2025

Oregon State University

Bachelor of Science in Mechanical Engineering | GPA: 3.92/4.0

Minors in Aerospace Engineering and Computer Science

Corvallis, OR

September 2022

SKILLS

Software: MATLAB, Python, ANSYS, OpenFOAM, FLOW3D, SolidWorks, OnShape

Hardware: Metal additive manufacturing, FDM 3D printing, Mechanical assembly, Machining

Analysis: Heat transfer, Computational fluid dynamics, Thermal imaging, Applied machine learning

PhD RESEARCH

Carnegie Mellon University

Thesis: *Understanding Melt Pool and Plume Dynamics in Laser Powder Bed*

Fusion Through Multiphysics Modeling and Two-Color Thermal Imaging

Spectral Observation and Simulation of Metal Vapor Plumes

Pittsburgh, PA

August 2022 – Present

- Analyzed multimodal spectral data on metallic vapor plumes to ascertain vapor interaction with high power laser irradiation and develop a physically based model of laser-based evaporation of metals
- Identified spectral signatures of titanium vapor atoms and vapor condensate regions with upwards of 20% light attenuation
- Creating a multiphysics CFD model of vapor plume evaporation with a custom nucleation submodel and laser-vapor interaction for use in quantification of laser beam attenuation and additive manufacturing process planning development

Hybrid Ratiometric and Illuminated Process Monitoring

- Co-developed a hybrid illuminated and ratiometric process monitoring technique to determine cooling rates in additive manufacturing processes by combining thermal gradients and morphological data obtained via laser reflections
- Developed a UNet machine learning model with >85% F1 Score to efficiently segment illuminated image video frames
- Reported temporally and spatially varying solidification cooling rates in 316L stainless steel melt pools between 170,000 and 3,000,000 K/s that demonstrate process phenomena and provide a foundation for process monitoring correlations

Simulated Imaging of L-PBF Melt Pools

- Created an algorithm to simulate ratiometric thermal imaging on reflecting concave surfaces using radiation heat transfer physics and imported CFD data for understanding high temperature thermal distributions in liquid metals
- Reproduced temperature distributions to within 4-6% of experimental values and provided imaging recommendations for experimentalists to use when imaging

Multiphysics CFD Melt Pool Simulation

- Developed and validated multiphysics CFD models of laser powder bed fusion (L-PBF) additive manufacturing in OpenFOAM® and FLOW3D® to study melt pool dynamics and evaporation
- Simulated L-PBF vapor plumes and found vapor velocities on the order of 100x greater than melt pool flow velocities

PUBLICATIONS

- Weeks, C. M.,** Myers, A. J., Malen, J. A., & Singh, S. (2025). Validation of OpenFOAM modeling of additive manufacturing melt pool dynamics against geometric and thermal experiments. *Journal of Manufacturing Processes*, 152, 237–249.
- Quirarte, G., Myers, A. J., Gourley, A., **Weeks, C. M.,** Reeja-Jayan, B., Beuth, J., & Malen, J. A. (2025). High speed thermal imaging and modeling of laser powder bed fusion manufactured WC–Ni cemented carbides. *Additive Manufacturing*, 110, 104913.
- Weeks, C. M.,** Malen, J. A., & Singh, S. (2025). Resolving Experimental Artifacts in Thermal Imaging of Laser Powder Bed Fusion Melt Pools. (in review)

CONFERENCES/PRESENTATIONS

- Weeks, Craig.** “Interpreting Temperature Distributions and Cooling Rates in Laser Powder Bed Fusion Through Surface Geometry and Illuminated Imaging.” Materials Science and Technology Technical Meeting and Exhibition, Columbus, OH, 1 October 2025, *Conference Presentation*
- Weeks, Craig.** “Simulation and Validation of Melt Pool Physics in the L-PBF Additive Manufacturing Process.” Materials Science and Technology Technical Meeting and Exhibition, Pittsburgh, PA, 8 October 2024, *Conference Presentation*
- Weeks, Craig.** “Numerical Simulation of Melt Pool Physics in Metal Additive Manufacturing Processes.” Solid Freeform Fabrication Symposium, Austin, TX, 13 August 2024, *Conference Presentation*
- Weeks, Craig.** “Numerical Simulation of Melt Pool Physics in Metal Additive Manufacturing Processes.” ASTFE Thermal and Fluids Engineering Conference, Corvallis, OR, 22 April 2024, *Conference Presentation*
- Weeks, Craig.** “Numerical Simulation of Melt Pool Physics in Metal Additive Manufacturing Processes.” Materials Science and Technology Technical Meeting and Exhibition, Columbus, OH, 2 October 2023, *Conference Presentation*
- Weeks, Craig.** “Hardware Design for an Efficient Inverter used in Electric Aircraft Power Systems.” Oregon Space Grant Consortium Student Symposium, Corvallis, OR, 13 November 2020, *Conference Presentation*

INTERNSHIP EXPERIENCE

NASA/HX5 – Cleveland, OH

- Electrical Systems Engineer Intern January 2022 – August 2022
- Designed mechanical packaging and integrated cooling elements of the 37.5kW electric power inverter in SolidWorks for the NASA SUSAN hybrid electric aircraft prototype
- Intern June 2021 – August 2021
- Implemented a Qt and C++ application for data acquisition and testing of electronic components on the X-57 Maxwell electric aircraft

TEACHING ASSISTANT EXPERIENCE

Carnegie Mellon University

Pittsburgh, PA

Lead Teaching Assistant / 24-321 Thermal Fluids Experimentation

January 2024 – May 2025

- Facilitated online office hours and assisted students outside of classroom hours
- Led undergraduate lab sessions and acted autonomously to respond to student questions and requests

AWARDS AND HONORS

National Defense Science and Engineering Graduate Fellow

April 2023

Oregon State University George and Evelyn Lundstrom Scholarship

September 2021

National Merit Scholar

March 2018

VOLUNTEER EXPERIENCE

Volunteer Instructor, CMU Gelfand Outreach Center – Pittsburgh, PA

July 2023 – July 2025

Internship Peer Mentor, NASA Glenn Research Center – Cleveland, OH

June 2020 – August 2021